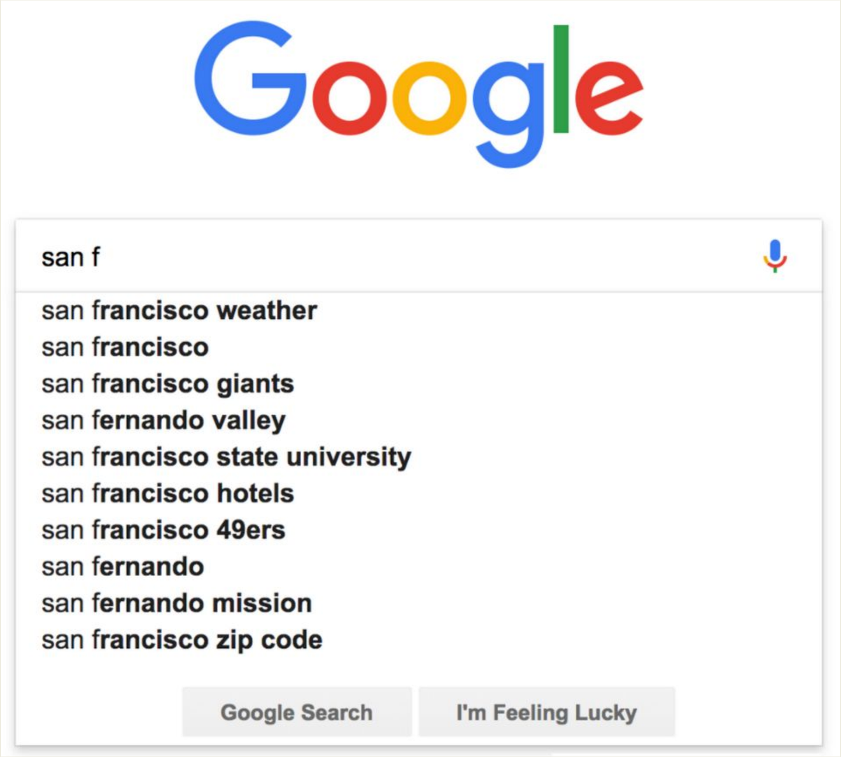
Sequence Learning – Recurrent Neural Network - LSTM

Sequence Learning Problems

- In feed-forward and convolutional neural networks the size of the input was always fixed.
- For example, we fed fixed size (32 x 32) images to convolutional neural networks for image classification.
- Each input to the network was independent of the previous or future inputs.
- The computations, outputs and decisions for two successive images are completely independent of each other.

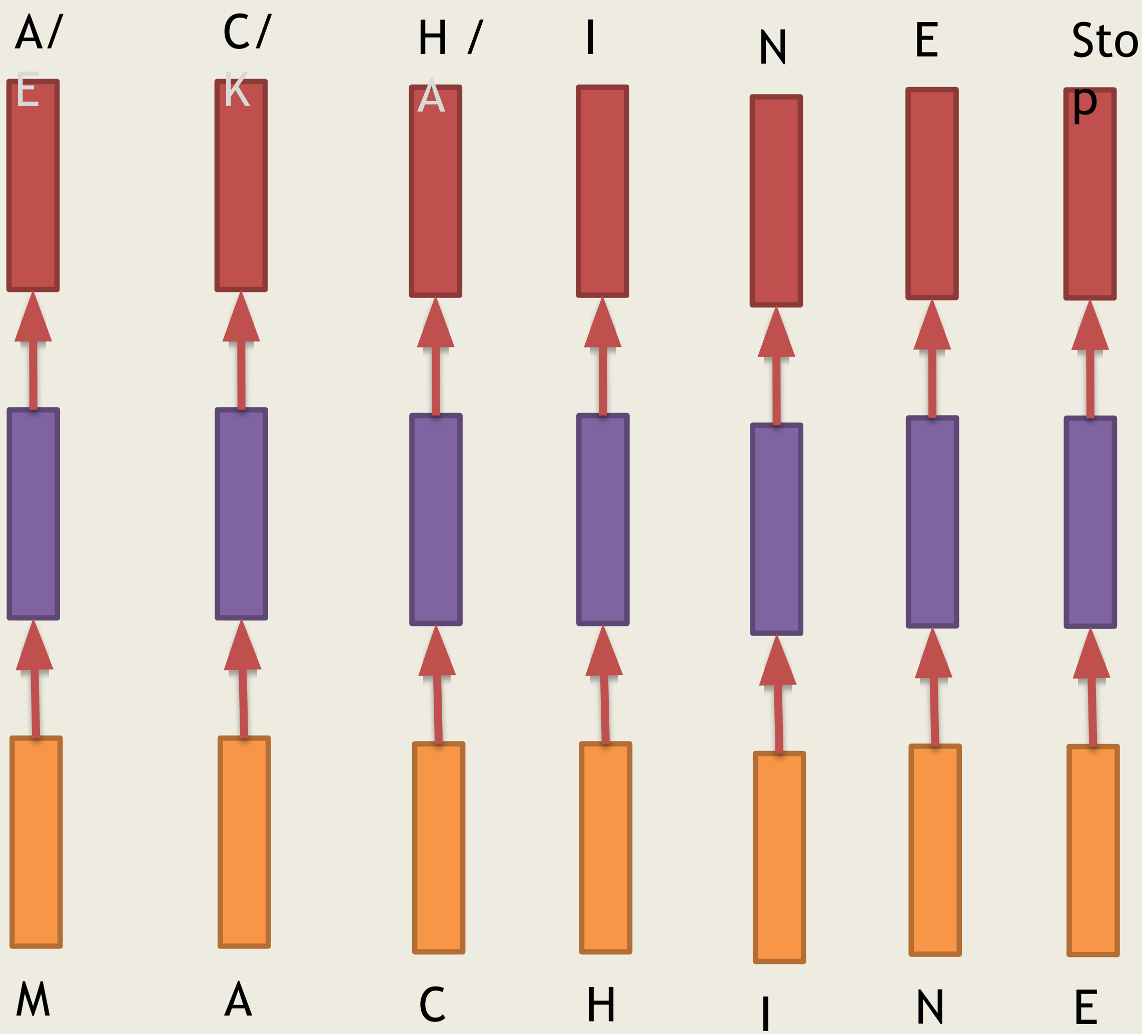
Sequence Learning Problems

- In many applications the input is not of a fixed size.
- Successive inputs may not be independent of each other.
- The length of the inputs and the number of predictions you need to make is not fixed.



Task of autocompletion

Sequence Learning Problems



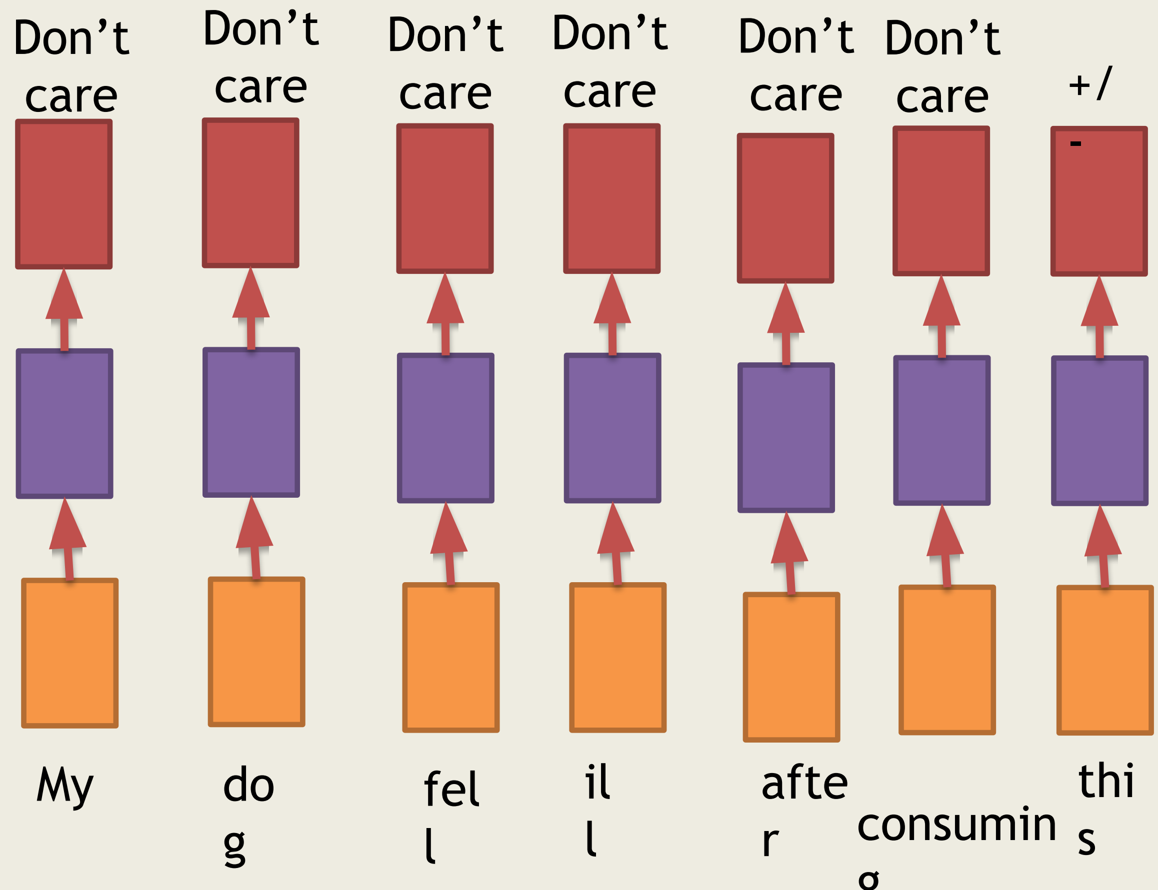
Each network is performing the task of input character output

Sequence Learning Problems

- To look at a sequence of (dependent) inputs and produce an output (or outputs)
- Each input corresponds to one time step
- Sometimes we may not be interested in producing an output at every stage
- Instead we would look at the full sequence and then produce an output

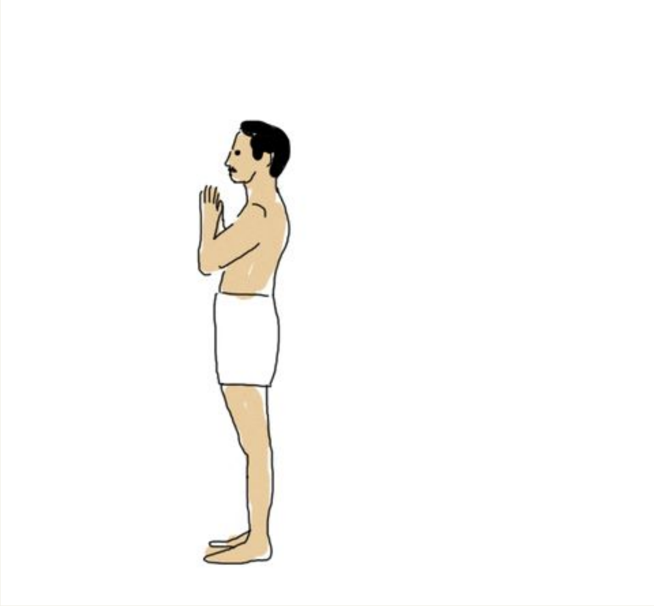
Sequence Learning Problems

- Consider the task of predicting the polarity of a dog food.



Sequence Learning Problems

- Sequences could be composed of any thing (not just words)



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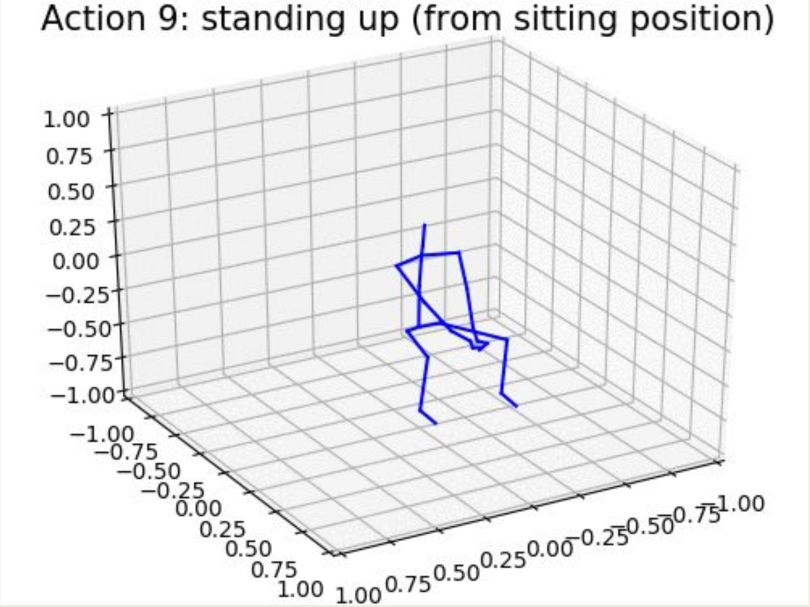
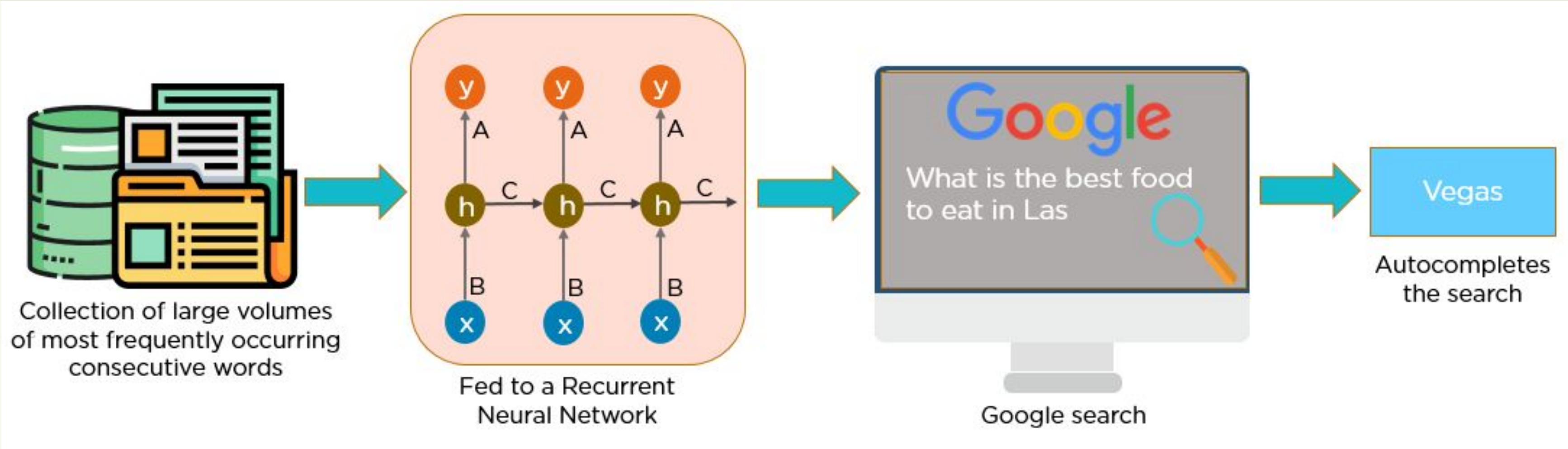


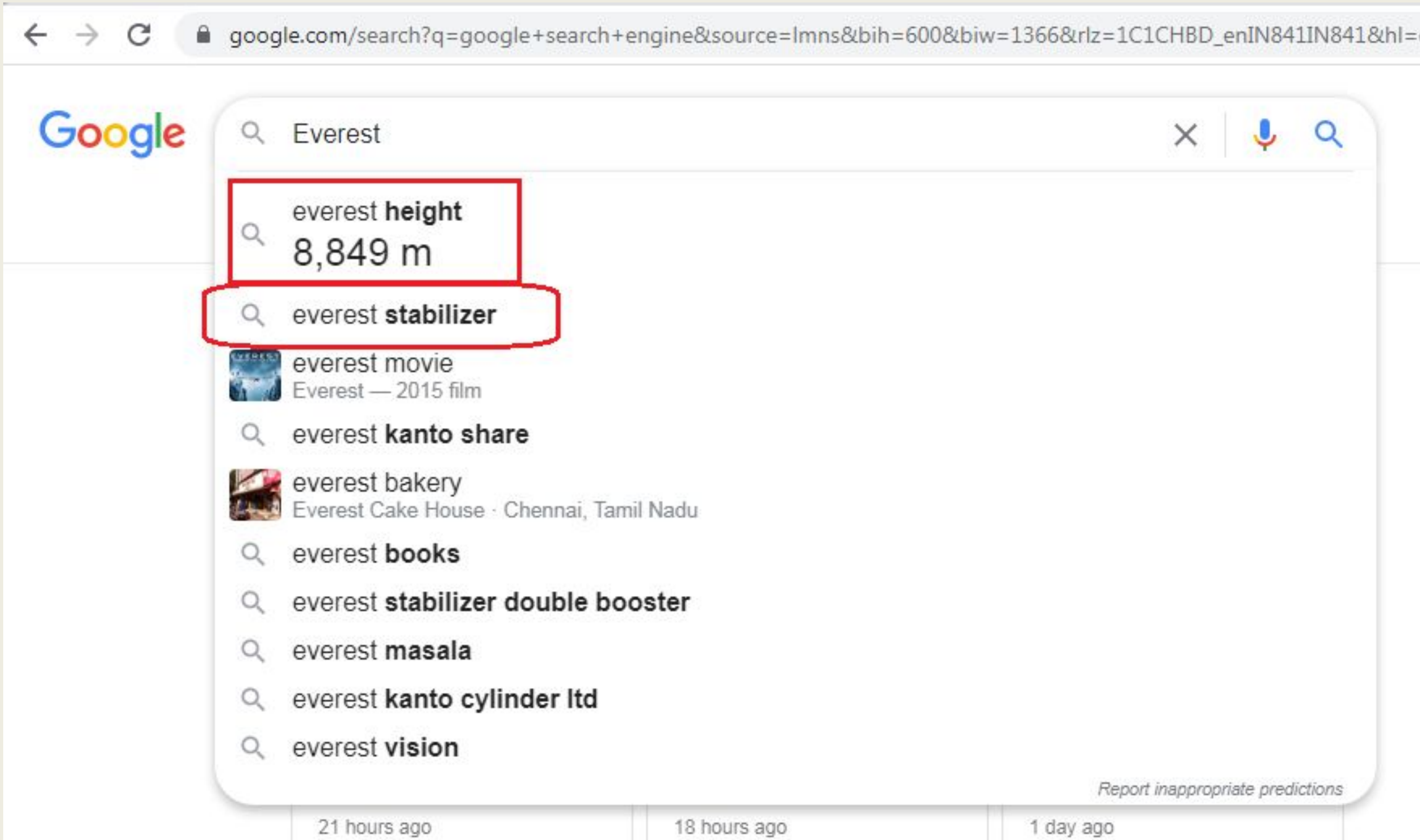
Human Activity
Recognition

Recurrent Neural Network

- A **recurrent neural network** (RNN) is a type of artificial **neural network**.
- Designed to recognize a data's sequential characteristics and use patterns to predict the next likely scenario.

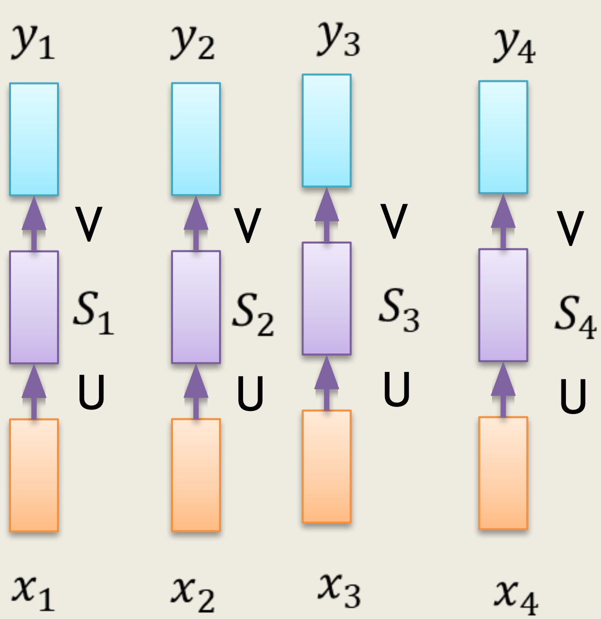


Recurrent Neural Network



Recurrent Neural Network

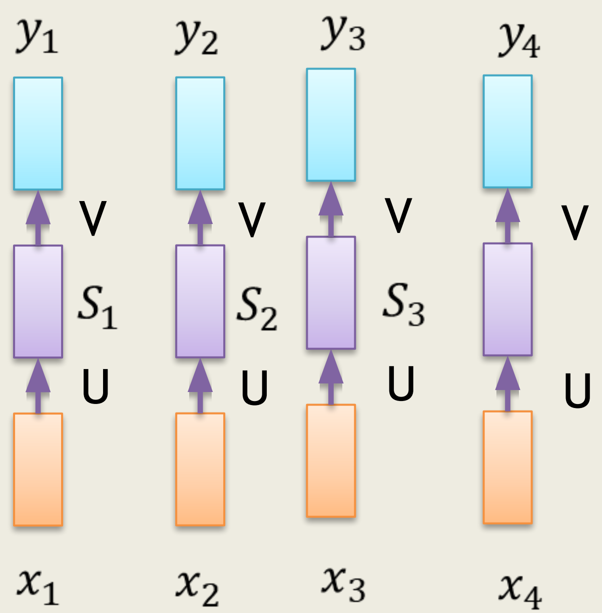
- Account for dependence between inputs
- Account for variable number of inputs
- Make sure that the function executed at each time step is the same



Since, we want the same function to be executed at each timestep we should share the same network (i.e., same parameters at each timestep)

Since we are simply going to compute the same function (with same parameters) at each timestep, the number of timesteps doesn't matter

Recurrent Neural Network



x_t is the input to the network at time step t

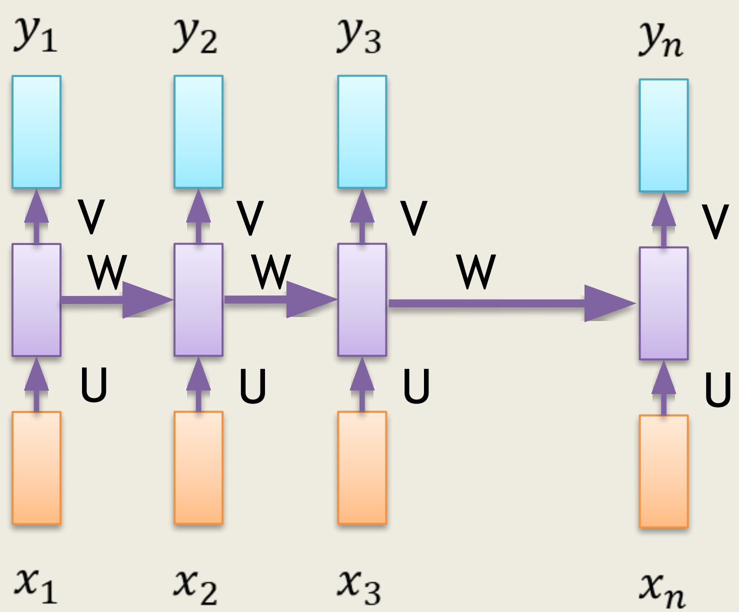
Example : one-hot vector corresponding to a word of a sentence.

Weights : parameterized by a weight matrix U, hidden-to-hidden recurrent connections parameterized by a weight matrix W, and hidden-to-output connections parameterized by a weight matrix V and all these weights *(U,V,W)* are shared across time.

Bias : c and b

Recurrent Neural Network

- Account for dependence



$$s_i = \sigma(Ux_i + Ws_{i-1} + b)$$

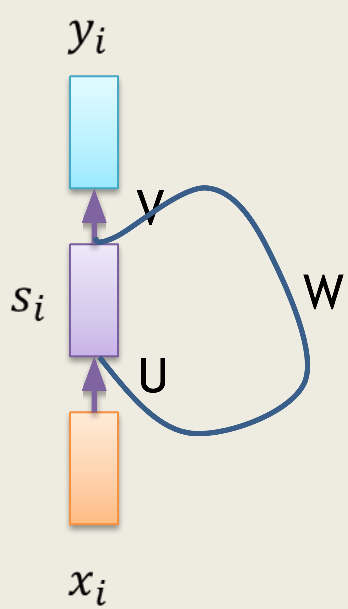
$$y_i = \mathcal{O}(Vs_i + c)$$

Where s_i is the state of the Network at timestep i

Parameters W, U, V, c, b are shared across timesteps

Recurrent Neural Network

- Account for dependence



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Thank You